

## 5.15

Create a function and then use central difference to calculate the derivative of the function in the range of  $-2 < x < 2$ .

$$f(x) = 1 + \frac{1}{2} \tanh(2x)$$

and the analytic derivative of that is:

$$f'(x) = \operatorname{sech}^2(2x) = \frac{1}{\cosh^2(2x)}$$

Python doesn't have `sech` built in but it does have `cosh` so we'll just use that form.

```
In [1]: import numpy as np
import matplotlib.pyplot as plt
```

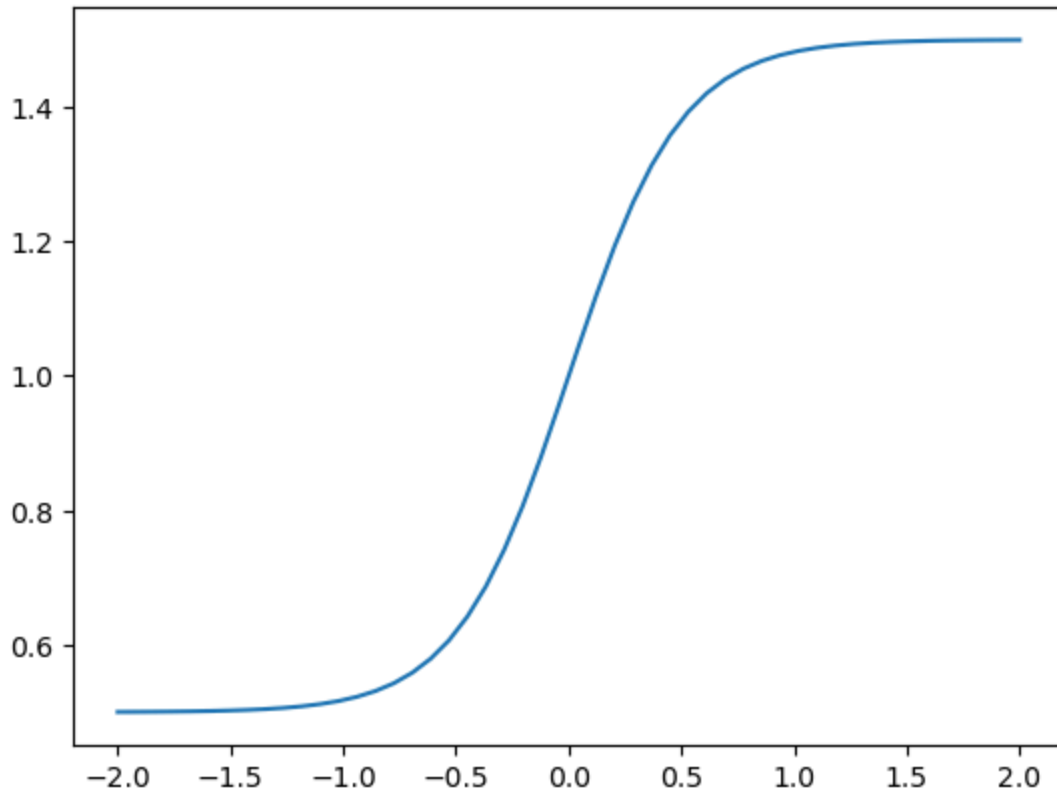
### First let's plot the function!

```
In [33]: def f(x):
          return 1+1/2*np.tanh(2*x)

x = np.linspace(-2,2)

# plot
fig, ax = plt.subplots()
ax.plot(x, f(x))
```

```
Out[33]: [<matplotlib.lines.Line2D at 0x784dad608950>]
```



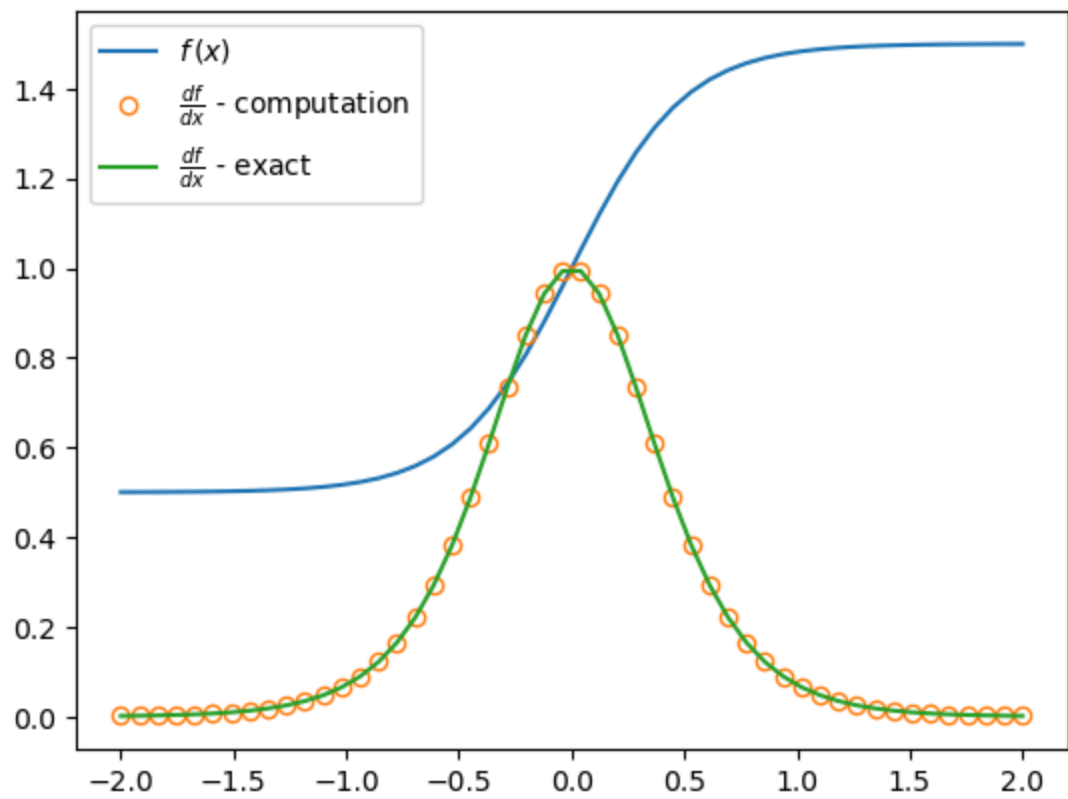
Now we'll use our midpoint derivative formula!

```
In [39]: def mid(function,x_value, stepSize):
          return (function(x_value+stepSize)-function(x_value-stepSize))/(2*stepSize)

          derivative = []
          for i in x:
              f_prime = mid(f, i, 0.001)
              derivative.append(f_prime)

          fig1, ax1 = plt.subplots()
          ax1.plot(x, f(x))
          ax1.plot(x, derivative, 'o', mfc='none')
          ax1.plot(x, 1/np.cosh(2*x)**2) # the exact answer
          ax1.legend([r'$f\,(x)$', r'$\frac{df}{dx}$ - computation', r'$\frac{df}{dx}$ - exact'])
```

Out[39]: <matplotlib.legend.Legend at 0x784dad9761e0>



In [ ]: